Project Proposal: PlantCare (EcoVision)

Intelligent Plant Disease Classification Platform

Group No. 10

Registration Numbers: E/22/032, E/22/124, E/22/232, E/22/237

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Course: CO543/CO5430 Computer Vision Project Assignment - 2026
Track: Application Track

1 Problem Statement and Motivation

Plant diseases significantly reduce crop yield and quality, posing a major threat to food security and agricultural productivity. Many farmers, particularly in rural areas, lack immediate access to agricultural experts, which leads to delayed diagnoses and ineffective treatments. The objective of this project is to develop an AI-powered computer vision platform that accurately detects and classifies plant diseases from leaf images. By automating this process, the platform will assist farmers in making timely decisions, reducing crop losses, and promoting sustainable farming practices.

2 Related Work

Recent advancements in deep learning have significantly improved the automation of agricultural monitoring. Previous works have extensively utilized Convolutional Neural Networks (CNNs) for leaf image classification. While many existing solutions provide basic classification, they often lack end-to-end integration with practical treatment recommendations for farmers. Our project builds upon the success of state-of-the-art architectures like ResNet and EfficientNet, adapting them specifically to our selected crop classes while providing a complete, deployable pipeline (mobile/web) that translates raw predictions into actionable agricultural advice.

3 Dataset

We will responsibly prepare and utilize established, publicly available datasets, primarily relying on the **PlantVillage Dataset**, **PlantDoc Dataset**, and the **Kaggle Plant Disease Dataset**.

- **Size and Classes:** The dataset includes multiple classes of healthy and diseased leaves, such as Apple Scab, Corn Rust, Tomato Early/Late Blight, Tomato Mosaic Virus, and Potato Early Blight.
- **Handling:** Data will be split rigorously into training, validation, and testing sets to ensure fair evaluation. Images will undergo preprocessing steps including resizing, normalization, and data augmentation to improve model generalization.

4 Planned Method and Baseline

The core of the project involves training an image classification model using PyTorch. The workflow includes image validation, preprocessing, deep learning inference, and generation of a confidence score.

- **Baseline Model:** As required by the assignment guidelines, our initial baseline will be a simple CNN architecture or a pretrained ResNet50 model without fine-tuning.
- **Improved/Adapted Method:** For our main technical contribution, we will implement and fine-tune advanced, lightweight architectures such as **EfficientNet-B0** or **MobileNetV3**. We will compare these against our baseline through ablation studies (e.g., evaluating with and without specific data augmentations).

5 Evaluation Metrics

To quantitatively evaluate our task, we will use the recommended metrics for image classification:

- **Accuracy, Precision, Recall, and F1-score:** To measure the overall correctness and balance of our predictions across different disease classes.
- **Confusion Matrix:** To analyze per-class performance and identify specific diseases the model struggles to differentiate.
- **Inference Time:** To ensure the model is efficient enough for integration into a FastAPI backend and mobile frontend.

We will also report qualitative visual evidence, discussing both successful classifications and failure cases.

6 Risks and Limitations

- **Data Bias and Generalizability:** Models trained on controlled datasets (like PlantVillage) often struggle with real-world, noisy backgrounds. We plan to mitigate this by mixing in datasets with varied backgrounds (like PlantDoc) and heavily utilizing data augmentation.
- **Computational Constraints:** Training deep models like EfficientNet requires significant resources. We will keep model sizes and batch sizes realistic for the available undergraduate computing platforms.
- **Integration Challenges:** Connecting the PyTorch AI model with the FastAPI backend and Next.js/Flutter frontends within the tight timeline could introduce bottlenecks. We will ensure the CV pipeline is developed and tested independently before full integration.

7 Timeline

The project timeline aligns with the fixed milestones of the course:

- **14 Jul 2026:** M1 - Proposal and project plan submission.
- **28 Jul 2026:** M2 - Dataset preparation, preprocessing, and baseline model implemented.
- **18 Aug 2026:** M3 - Working prototype of the fine-tuned method with preliminary metrics and failure case analysis.
- **25 Aug 2026:** M4 - Experiment freeze; final results collected. Focus shifts to analysis and report writing.
- **1 Sep 2026:** M5 - Draft report structure, repository cleanup, and demo checklist preparation.
- **7 Sep 2026:** M6 - Final submission (IEEE format report, GitHub repository, and live demonstration).